

HDOC DAY-17 ACTIVITY

QUESTION 1 – Menu-Driven Digit Operations with User-Defined Functions

Objective: Write a menu-driven C program using switch-case to perform four digit-based operations on a positive integer. Each menu option is implemented using one separate user-defined function with void return type and a simple function call.

- Generate the largest possible number using its digits
- Generate the smallest possible number using its digits.
- Swap its first and last digits.
- Check whether it is a palindrome number.

1. Problem Understanding

- **Input:** A positive integer entered by the user inside the function corresponding to the selected menu option.
- **Output:** The result of the selected digit operation.
- **Functions:** A separate void function is used for each menu option. Each function reads its own input, processes it, and displays the result.
- **Operations:** Generate the largest possible number using the digits; generate the smallest possible number using the digits; swap the first and last digits; check whether the number is a palindrome.

Operation Summary

Choice	Operation	Main Logic
1	Largest possible number	Extract the digits, sort them in descending order, and display the resulting number.
2	Smallest possible number	Extract the digits, sort them in ascending order, and place the first non-zero digit first so the result remains a valid number.
3	Swap first and last digits	Find the first and last digits, replace them, and display the resulting number.
4	Palindrome check	Compare the original number with its reverse. If both are equal, it is a palindrome.
Other	Invalid choice	Display an appropriate invalid-choice message.

2. Standard Test Cases

Test Case	Choice	Input	Expected Output	Actual Output	Result
1	1	n = 58321	Largest possible number = 85321	85321	Pass
2	2	n = 58321	Smallest possible number = 12358	12358	Pass
3	3	n = 12345	After swapping = 52341	52341	Pass
4	4	n = 1221	It is a palindrome number	It is a palindrome number	Pass

5	4	n = 12345	It is not a palindrome number	It is not a palindrome number	Pass
6	2	n = 10230	Smallest possible number = 10203	10203	Pass
7	3	n = 7	After swapping = 7	7	Pass
8	9	—	Invalid choice	Invalid choice	Pass

3. Algorithm

A. Common Menu Logic

1. Start the program.
2. Display the menu containing the four operations.
3. Read the user's choice.
4. Use switch-case to call the corresponding void function.
5. The selected function reads the positive integer, processes it, and displays the result.
6. If the choice is invalid, display an invalid-choice message.
7. Stop the program.

B. Case 1 – Generate the Largest Possible Number

1. Read the positive integer.
2. Extract each digit and store the digits in an array.
3. Sort the digits in descending order.
4. Print the digits from left to right to form the largest possible number.
5. Display the result.

C. Case 2 – Generate the Smallest Possible Number

1. Read the positive integer.
2. Extract each digit and store the digits in an array.
3. Sort the digits in ascending order.
4. If the first digit is zero, find the first non-zero digit and place it first.
5. Print the remaining digits in ascending order.
6. Display the result.

D. Case 3 – Swap First and Last Digits

1. Read the positive integer.
2. Find the last digit using $n \% 10$.
3. Find the first digit and the place value of the first digit.
4. Remove the first and last digit positions from the number.
5. Exchange the first and last digits and reconstruct the number.
6. Display the result.

E. Case 4 – Check Palindrome Number

1. Read the positive integer.
2. Store the original number.
3. Reverse the number by repeatedly extracting the last digit.
4. Compare the reversed number with the original number.
5. If they are equal, display that the number is a palindrome; otherwise display that it is not a palindrome.

4. C Program

```
#include <stdio.h>

void generateLargest(void)
{
    int n, temp, digit[20], count = 0;
    int i, j, swap;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    temp = n;
    while (temp > 0)
    {
        digit[count++] = temp % 10;
        temp = temp / 10;
    }

    for (i = 0; i < count - 1; i++)
    {
        for (j = i + 1; j < count; j++)
        {
            if (digit[i] < digit[j])
```

```

        {
            swap = digit[i];
            digit[i] = digit[j];
            digit[j] = swap;
        }
    }
}

printf("Largest possible number = ");
for (i = 0; i < count; i++)
    printf("%d", digit[i]);
printf("\n");
}

```

```

void generateSmallest(void)
{
    int n, temp, digit[20], count = 0;
    int i, j, swap, firstNonZero;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    temp = n;
    while (temp > 0)
    {
        digit[count++] = temp % 10;
        temp = temp / 10;
    }

    for (i = 0; i < count - 1; i++)
    {
        for (j = i + 1; j < count; j++)
        {
            if (digit[i] > digit[j])
            {
                swap = digit[i];
                digit[i] = digit[j];
                digit[j] = swap;
            }
        }
    }

    if (count > 1 && digit[0] == 0)
    {
        firstNonZero = 1;
    }
}

```

```

while (firstNonZero < count && digit[firstNonZero] == 0)
    firstNonZero++;

swap = digit[0];
digit[0] = digit[firstNonZero];
digit[firstNonZero] = swap;
}

printf("Smallest possible number = ");
for (i = 0; i < count; i++)
    printf("%d", digit[i]);
printf("\n");
}

void swapFirstLast(void)
{
    int n, temp, first, last, divisor = 1;
    int middle, result;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n < 10)
    {
        printf("After swapping first and last digits = %d\n", n);
        return;
    }

    last = n % 10;
    temp = n;

    while (temp >= 10)
    {
        temp = temp / 10;
        divisor = divisor * 10;
    }

    first = temp;
    middle = (n % divisor) / 10;
    result = last * divisor + middle * 10 + first;

    printf("After swapping first and last digits = %d\n", result);
}

void checkPalindrome(void)

```

```

{
    int n, temp, digit, reverse = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    temp = n;
    while (temp > 0)
    {
        digit = temp % 10;
        reverse = reverse * 10 + digit;
        temp = temp / 10;
    }

    if (n == reverse)
        printf("It is a palindrome number.\n");
    else
        printf("It is not a palindrome number.\n");
}

int main(void)
{
    int choice;

    printf("MENU\n");
    printf("1. Generate largest possible number using its digits\n");
    printf("2. Generate smallest possible number using its digits\n");
    printf("3. Swap first and last digits\n");
    printf("4. Check whether it is a palindrome number\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            generateLargest();
            break;

        case 2:
            generateSmallest();
            break;

        case 3:
            swapFirstLast();
            break;
    }
}

```

case 4:

```
    checkPalindrome();  
    break;
```

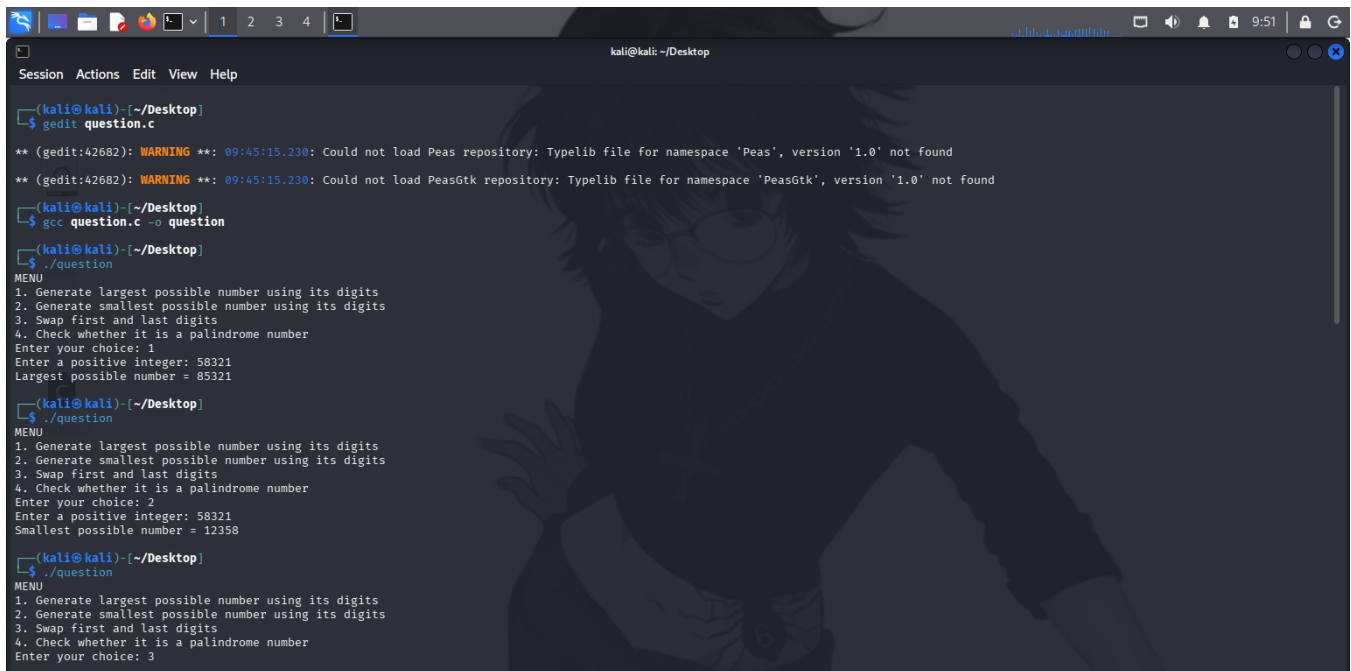
default:

```
    printf("Invalid choice.\n");  
}
```

```
return 0;
```

```
}
```

5. Compilation and Execution Evidence



The screenshot shows a terminal window titled 'kali@kali: ~/Desktop'. The user has created a file 'question.c' and compiled it with 'gcc question.c -o question'. The program is then executed with './question'. The program displays a menu with four options: 1. Generate largest possible number using its digits, 2. Generate smallest possible number using its digits, 3. Swap first and last digits, and 4. Check whether it is a palindrome number. The user enters choice 1 and a positive integer 58321, resulting in the largest possible number 85321. The user then enters choice 2 and the same integer, resulting in the smallest possible number 12358. Finally, the user enters choice 3, and the program prompts for the next choice.

```
kali@kali: ~/Desktop  
$ gedit question.c  
  
** (gedit:42682): WARNING **: 09:45:15.230: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found  
** (gedit:42682): WARNING **: 09:45:15.230: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found  
  
kali@kali: ~/Desktop  
$ gcc question.c -o question  
  
kali@kali: ~/Desktop  
$ ./question  
MENU  
1. Generate largest possible number using its digits  
2. Generate smallest possible number using its digits  
3. Swap first and last digits  
4. Check whether it is a palindrome number  
Enter your choice: 1  
Enter a positive integer: 58321  
Largest possible number = 85321  
  
kali@kali: ~/Desktop  
$ ./question  
MENU  
1. Generate largest possible number using its digits  
2. Generate smallest possible number using its digits  
3. Swap first and last digits  
4. Check whether it is a palindrome number  
Enter your choice: 2  
Enter a positive integer: 58321  
Smallest possible number = 12358  
  
kali@kali: ~/Desktop  
$ ./question  
MENU  
1. Generate largest possible number using its digits  
2. Generate smallest possible number using its digits  
3. Swap first and last digits  
4. Check whether it is a palindrome number  
Enter your choice: 3
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
Enter a positive integer: 12345
After swapping first and last digits = 52341

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Generate largest possible number using its digits
2. Generate smallest possible number using its digits
3. Swap first and last digits
4. Check whether it is a palindrome number
Enter your choice: 4
Enter a positive integer: 1221
It is a palindrome number.

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Generate largest possible number using its digits
2. Generate smallest possible number using its digits
3. Swap first and last digits
4. Check whether it is a palindrome number
Enter your choice: 4
Enter a positive integer: 12345
It is not a palindrome number.

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Generate largest possible number using its digits
2. Generate smallest possible number using its digits
3. Swap first and last digits
4. Check whether it is a palindrome number
Enter your choice: 2
Enter a positive integer: 10230
Smallest possible number = 10023

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Generate largest possible number using its digits
2. Generate smallest possible number using its digits
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
2. Generate smallest possible number using its digits
3. Swap first and last digits
4. Check whether it is a palindrome number
Enter your choice: 4
Enter a positive integer: 12345
It is not a palindrome number.

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Generate largest possible number using its digits
2. Generate smallest possible number using its digits
3. Swap first and last digits
4. Check whether it is a palindrome number
Enter your choice: 2
Enter a positive integer: 10230
Smallest possible number = 10023

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Generate largest possible number using its digits
2. Generate smallest possible number using its digits
3. Swap first and last digits
4. Check whether it is a palindrome number
Enter your choice: 3
Enter a positive integer: 7
After swapping first and last digits = 7

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Generate largest possible number using its digits
2. Generate smallest possible number using its digits
3. Swap first and last digits
4. Check whether it is a palindrome number
Enter your choice: 9
Invalid choice.

(kali@kali)-[~/Desktop]
$
```

6. Verification of All Test Cases

Test Case	Choice	Input	Expected Output	Actual Output	Result
1	1	n = 58321	Largest possible number = 85321	Largest possible number = 85321	Pass

2	2	n = 58321	Smallest possible number = 12358	Smallest possible number = 12358	Pass
3	3	n = 12345	After swapping = 52341	After swapping = 52341	Pass
4	4	n = 1221	It is a palindrome number	It is a palindrome number	Pass
5	4	n = 12345	It is not a palindrome number	It is not a palindrome number	Pass
6	2	n = 10230	Smallest possible number = 10203	Smallest possible number = 10203	Pass
7	3	n = 7	After swapping = 7	After swapping = 7	Pass
8	9	—	Invalid choice	Invalid choice	Pass